Marketability And Economic Value

AEGIS

Marketability And Economic Value

The Robo Rescue Spider is not intended to be limited to a university robotics demonstration or a single search-and-rescue application. The broader concept is to develop an adaptable robotic platform capable of operating in environments where obtaining information may be difficult, expensive or hazardous for humans.

The market opportunity is based on a simple economic principle:

If a robot can collect information or perform an inspection without immediately exposing a person to the same hazard, the robot can create value through improved safety, reduced exposure, better decision-making and potentially reduced operational costs.

The proposed system therefore has potential applications across emergency response, mining, industrial inspection, infrastructure, energy and disaster management.

The marketability of the Robo Rescue Spider is supported by three major observations.

First, South Africa has industries where hazardous environments represent a significant operational and economic concern. Mining alone contributed approximately R450.5 billion directly to South Africa's GDP in 2024, representing 6.1% of GDP, while providing employment to approximately 474,736 people.

Second, occupational risk remains a real issue. South African mines recorded 42 occupational fatalities and 1,841 occupational injuries in 2024. Although these figures represent significant improvements compared with previous years, they demonstrate that hazardous working environments remain a real problem.

Third, the global robotics industry is expanding. According to the International Federation of Robotics, 542,000 industrial robots were installed worldwide in 2024, more than twice the number installed ten years earlier . Professional service robots also reached almost 200,000 units sold in 2024, demonstrating that robotics is expanding beyond traditional factory automation .

These trends create an opportunity for a specialised robotic system focused on hazardous and difficult-to-access environments.

1.1 Target Industries

The Robo Rescue Spider has been designed around a core capability rather than a single industry: remote access and information gathering in environments that may be difficult or hazardous for humans.

This allows the same underlying platform to be adapted for different industries.

1.2 Emergency and Search & Rescue

The most direct application is emergency response.

The robot could be deployed following:

- Building collapses.
- Industrial accidents.
- Fires.
- Earthquakes.
- Landslides.
- Tunnel failures.
- Mining accidents.
- Other hazardous incidents.

Its primary role would be reconnaissance rather than replacing human responders.

A responder could use the robot to obtain visual information, investigate a pathway, identify obstacles or inspect an area before deciding whether personnel should enter.

This application is strongly aligned with existing emergency-response robotics research. NIST identifies collapsed structures, fires, industrial accidents, victim searching, hazard identification and situational awareness as applications for emergency-response robots.

1.3 Mining

Mining represents one of the strongest potential markets for the Robo Rescue Spider, particularly in South Africa.

South Africa's mining industry generated approximately R450.5 billion in direct GDP contribution in 2024, employed approximately 474,736 people, and generated approximately R585.6 billion in mineral exports .

The industry also faces environments where access can be difficult and dangerous.

In 2024, South African mining recorded:

- 42 occupational fatalities.
- 1,841 occupational injuries.
- 13 fatalities associated with fall-of-ground accidents.
- This creates a potential use case for robotic systems capable of inspecting areas before personnel enter them.

Possible mining applications include:

- Underground reconnaissance.
- Post-accident inspection.
- Inspection of unstable areas.
- Mapping difficult-to-access spaces.
- Camera-based environmental assessment.
- Infrastructure inspection.
- Remote investigation following incidents.

The opportunity is particularly relevant because improving safety does not only have social value. It also has economic value because safer operations can reduce disruption, protect skilled workers and support operational continuity.

1.4 Industrial Inspection

Factories, processing plants, warehouses and industrial facilities contain equipment and infrastructure that periodically require inspection.

A robotic platform could potentially be used to inspect areas that are:

- Confined.
- Difficult to reach.
- Hot.
- Structurally unsafe.
- Contaminated.
- Located around operating machinery.

Instead of sending a person immediately into the area, an organisation could first use the robot to obtain visual or sensor information.

This changes the robot from being purely a “rescue robot” into a broader inspection and reconnaissance platform.

1.6 Oil, Gas and Energy

Energy infrastructure can contain hazardous environments where remote inspection may provide value.

Potential applications include:

- Industrial plant inspection.
- Pipeline environments.
- Energy facilities.
- Storage facilities.
- Confined infrastructure.
- Post-incident assessment.

The value comes from obtaining information while reducing unnecessary human exposure.

The platform could also be adapted by changing its sensor payload depending on the industry.

1.7 Infrastructure Inspection

The same concept can potentially be applied to:

- Bridges.
- Tunnels.
- Construction sites.
- Drainage systems.
- Dam infrastructure.
- Underground infrastructure.
- Disaster-damaged buildings.

The World Bank and Development Bank of Southern Africa have highlighted the importance of infrastructure investment and resilience in South Africa, including the need to consider exposure of infrastructure assets to natural hazards [6].

A mobile inspection robot could therefore form part of a broader infrastructure-monitoring ecosystem.

1.8 Agriculture and Environmental Monitoring

Although this is not the project's primary market, the platform could eventually be adapted for environmental applications.

With suitable sensors, the robot could potentially assist with:

- Difficult terrain inspection.
- Environmental monitoring.
- Agricultural field assessment.
- Remote observation.
- Data collection.

This demonstrates the advantage of developing a modular platform rather than a single-purpose machine.

1.9 South African Market.

South Africa is an important starting market because the country combines a significant mining sector, industrial infrastructure, emergency-response requirements and a growing need for technological innovation.

Why South Africa?

South Africa does not need to create a completely artificial market for this technology. Several existing industries already spend money managing environments where safety, access and inspection are important.

The mining sector provides the clearest example.

According to the Department of Mineral and Petroleum Resources, mining contributed R450.5 billion directly to GDP in 2024, representing 6.1% of South Africa's GDP.

The sector also employed approximately 474,736 people and generated approximately R585.6 billion in mineral exports .

This means that mining is not simply a possible customer for Robo Rescue Spider. It is a strategically important part of the South African economy.

At the same time, safety remains a major concern.

The South African government reported 42 mining fatalities in 2024, even though this was the lowest number ever recorded and represented a 24% improvement from 55 fatalities in 2023.

The improvement should not be interpreted as evidence that the problem has disappeared. Instead, it demonstrates that safety interventions can produce measurable improvements and that the industry continues to pursue the goal of zero harm.

The Robo Rescue Spider could therefore be positioned as one additional technological tool within the broader safety strategy, rather than claiming that robotics alone can solve mining accidents.

South African economic opportunity

The opportunity extends beyond selling robots.

Developing the Robo Rescue Spider locally could create an ecosystem involving:

Robotics → Software → Electronics → Manufacturing → Sensors → Maintenance → Training → Data services

This is important because the economic value of the project could come from more than the physical robot.

A successful South African robotics company could create demand for:

- Robotics engineers.
- Software developers.
- Mechanical designers.
- Electrical engineers.
- Data scientists.
- Technicians.
- Manufacturing workers.
- Field operators.
- Robotics trainers.

Therefore, the project could contribute to the development of local technical skills while creating a product capable of being exported.

1.10 Global Market

The global market opportunity is based on the wider growth of robotics.

The International Federation of Robotics reported that 542,000 industrial robots were installed globally in 2024, more than twice the number installed ten years earlier .

This demonstrates that organisations around the world are increasingly adopting robotic systems.

Importantly, robotics is no longer restricted to traditional factory automation.

The IFR reported that almost 200,000 professional service robots were sold in 2024, representing 9% growth compared with the previous year .

This is important for Robo Rescue Spider because it belongs closer to the professional service robotics category than traditional manufacturing robotics.

The potential global markets include:

Africa

- Mining.
- Disaster management.
- Infrastructure inspection.
- Industrial inspection.
- Energy.
- Agriculture.
- Security and emergency services.

Europe

- Infrastructure inspection
- Industrial inspection
- Emergency response
- Energy.
- Construction
- Hazardous-environment operations.

North America

- Search and rescue.
- Mining.
- Industrial inspection.
- Disaster response.
- Infrastructure.
- Energy.
- Public safety.

Asia

Asia represents the world's largest robotics market. The IFR reported that 74% of new industrial robot installations in 2024 occurred in Asia.

Rather than attempting to compete immediately with large-scale industrial robotics manufacturers, Robo Rescue Spider could focus on a specialised niche: mobile robotics for hazardous, irregular and difficult-to-access environments.

1.11 Economic Value Proposition

The economic value of Robo Rescue Spider is not simply the price of the robot.

The value comes from what the robot can potentially help an organisation avoid, improve or accomplish.

The proposed economic value can be understood through five areas.

1. Reducing human exposure

The strongest value proposition is safety.

If a robot can perform an initial inspection before a person enters a hazardous area, the organisation may reduce unnecessary exposure to danger.

The robot does not eliminate risk. Instead, it can potentially help humans make better decisions about when and where to enter.

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2. Improving decision-making

Information has economic value.

A responder who has information about an environment can potentially make a better decision than a responder operating with limited information.

The robot can therefore provide:

Observation → Information → Situational awareness → Better decision

This is particularly important in emergency situations where decisions must be made quickly.

3. Reducing unnecessary deployment

Not every dangerous-looking environment necessarily requires immediate human entry.

A robotic reconnaissance system could potentially determine whether an area requires further intervention.

This could help organisations use personnel and equipment more efficiently.

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4. Reducing downtime

In industrial environments, an incident or inspection can interrupt operations.

A robotic inspection platform could potentially collect information without requiring the same level of human access or preparation.

The economic benefit would depend on the particular industry and application, but the potential value comes from shortening or improving the information-gathering stage of an operation.

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5. Creating a local technology industry

The long-term economic value extends beyond the robot itself.

If Robo Rescue Spider becomes a commercial product, South Africa could potentially participate in the development, manufacturing and servicing of specialised robotics rather than simply importing robotic systems.

This aligns with a larger opportunity for Africa.

A World Bank/Rise Partnership report highlights Africa's substantial energy-transition mineral resources and argues that the continent has an opportunity to participate in global value chains, while also noting challenges involving infrastructure, skills and sustainable production.

This creates an interesting strategic connection:

Africa supplies important minerals → mining creates economic value → mining requires safer and more efficient technology → African companies can develop technology for African operating conditions → that technology can then be exported globally.

That is a much stronger economic story than simply saying:

“We want to sell a robot.”

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1. Potential Customers

The potential customer is not necessarily the person who physically operates the robot.

The customer is the organisation that has a financial or operational reason to use the technology.

Potential customers include:

Primary customers

Mining companies

They could use the system for underground reconnaissance, inspection and post-incident assessment.

Emergency-response organisations.

Fire and rescue services, disaster-management organisations and search-and-rescue teams could use the robot for hazardous-area reconnaissance.

Industrial companies.

Manufacturing and processing companies could use the platform for inspection of difficult-to-access areas.

Engineering and inspection companies

Specialised inspection companies could operate the robot as a service for multiple clients.

Secondary customers

Potential future customers could include

- Infrastructure companies.
- Energy companies.
- Construction companies.
- Environmental organisations.
- Research institutions.
- Universities.

- Government agencies.
- International humanitarian organisations.

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5 Business / Revenue Model

The business model should not depend only on selling the physical robot.

A stronger model is a Robotics-as-a-Service (RaaS) approach.

Instead of saying:

“Buy our robot.”

the company could offer:

“We provide robotic inspection and reconnaissance as a service.”

This model could include:

Model 1: Direct robot sales

The customer purchases:

Robot + controller + sensors + software

This model is appropriate for large organisations that have their own technical teams.

Model 2: Robotics-as-a-Service

The customer pays for the use of the robot rather than purchasing the entire system.

The service could include:

Robot + trained operator + deployment + data collection + reporting

This could be particularly attractive for smaller organisations that cannot afford to purchase and maintain a specialised robot.

Model 3: Inspection contracts

The company could provide periodic robotic inspections.

For example:

“Quarterly underground robotic inspection.”

The customer pays a recurring fee for the service.

Model 4: Software and data services

The physical robot could be combined with software that provides:

Remote control;

Video monitoring;

Mapping;

Sensor analysis;

Data storage;

Inspection reports; and

Historical comparison.

This creates recurring revenue beyond the initial hardware sale.

Model 5: Customisation

Different industries have different requirements.

The same base platform could therefore be customised with different payloads.

For example:

Mining configuration → camera + environmental sensors

Fire configuration → thermal camera + gas sensors

Industrial configuration → inspection camera + specialised sensors

This creates additional revenue through hardware, software and integration.

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